AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION 2010

Physics Paper II

Time allowed: 2 hours 25 minutes Marks 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL EIGHT questions. Questions 6, 7 & 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

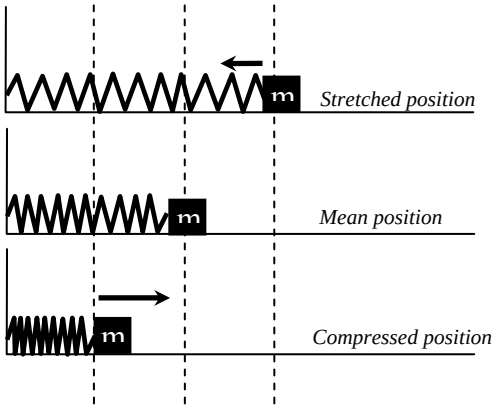
   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

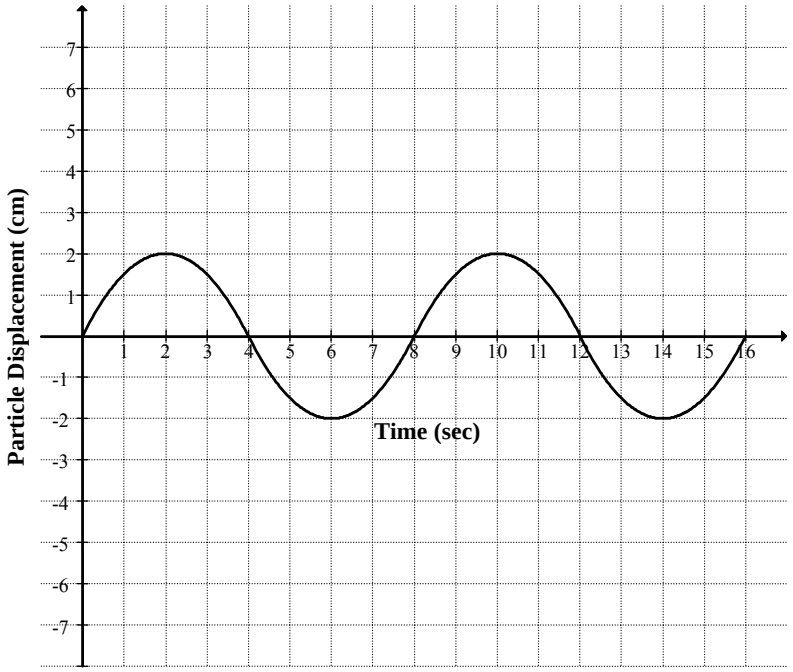
5. You may use a simple calculator if you wish.

Q.1. (Total 5 Marks)

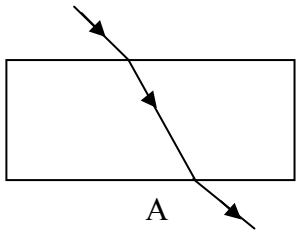
a. In the given figure, a body is attached with a spring. Recognize the motion and develop an expression of Hook's law. (3 Marks)

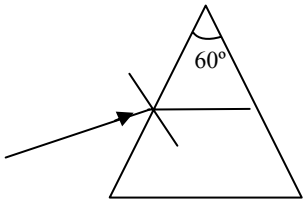


b. In the same diagram, draw another transverse wave travelling through the same medium with twice the amplitude and frequency. (2 Marks)

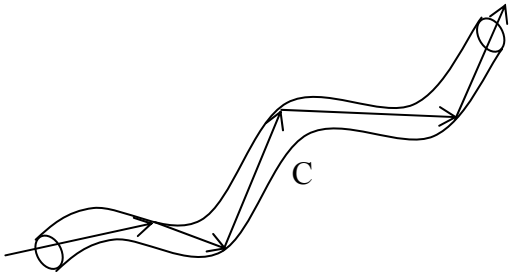


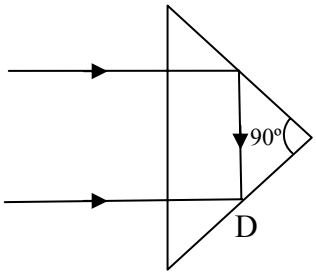
Q.2. (Total 4 Marks)


A



B


C



D

a. Which of the above diagrams shows total internal reflection? (1 Mark)

_____

b. What are the TWO conditions that are necessary for total internal reflection? (2 Marks)

_____

_____

_____

_____

c. Find the refractive index of water if its critical angle is 45°. (1 Mark)

_____

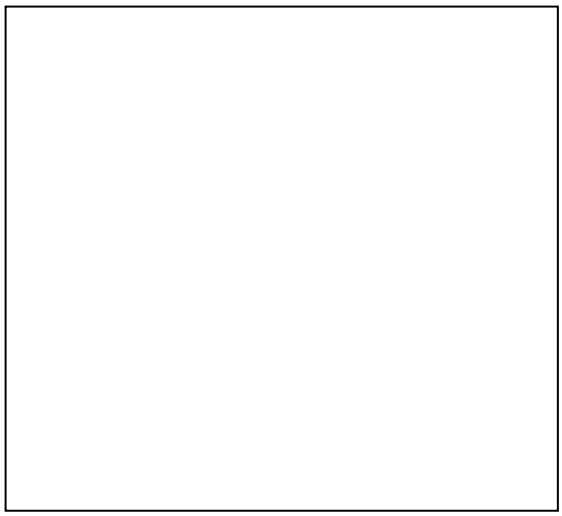
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Q.3. (Total 4 Marks)

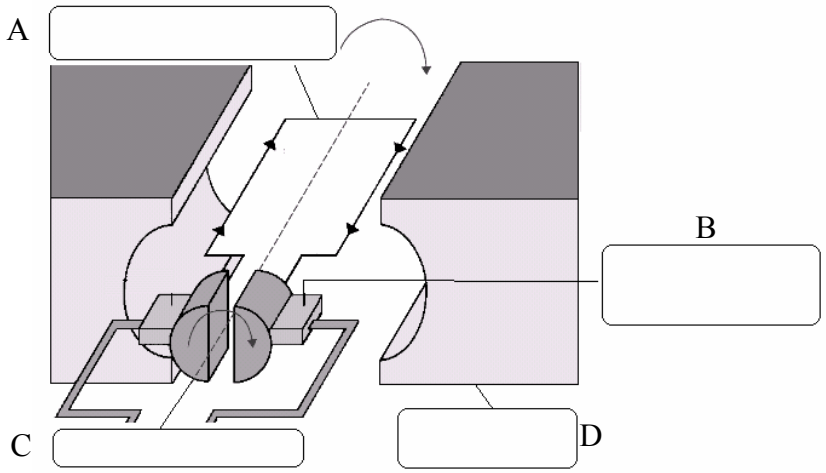
Two resistors of 18 ohms and 36 ohms are connected in parallel with a 24 volt battery. An ammeter is also placed in series to read the total current passing through the circuit.

a. Draw a diagram of this circuit. (1 Mark)



b. Calculate the equivalent resistance of the circuit. (3 Marks)

Q.4. (Total 8 Marks)



a. Recognize and label the above diagram in the given spaces. (5 Marks)

b. Briefly describe its working principle. (3 Marks)

Q.5. (Total 4 Marks)

a. Name a semi-conductor element. (1 Mark)

_____

b. How does the conducting ability of a semi-conductor compare with that of a metallic conductor? (2 Marks)

_____

_____

_____

_____

c. What is the effect on a pure semi-conductor after doping, so that it becomes an n-type material? (1 Mark)

_____

_____

Q.6. (Total 5 Marks)

EITHER

a. Reflection, refraction, interference and diffraction are the properties of waves. With the help of a ripple tank, discuss any TWO of them and draw any ONE diagram.

OR

b. The depth of an ocean is measured by using ultrasonic. Explain how it works.

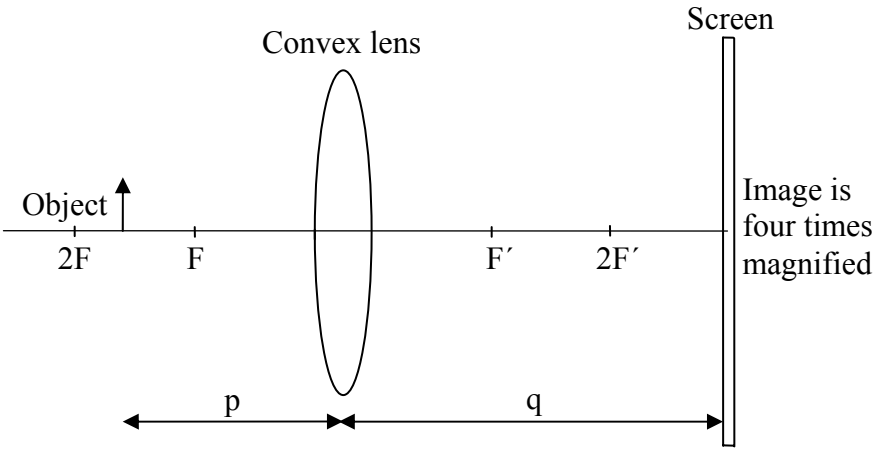
Q.7. (Total 5 Marks)

EITHER

a. A student finds that in order to read white board, he/she has to move closer to the board although he/she can read books very easily. Write the name and cause of the defect with ray diagram and how it can be removed. Draw a ray diagram related to the removing of this defect.

OR

b. Complete the given ray diagram and calculate the value of p, q and f, if p + q = 50 cm.



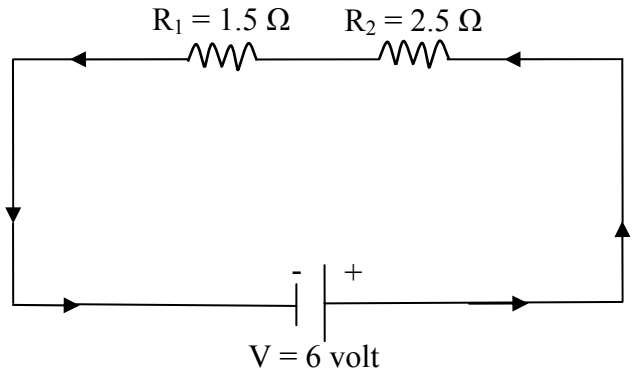
Q.8. (Total 5 Marks)

EITHER

a. We often hear that during a thunder-storm, buildings are struck by lightning; this is a hazard of electrostatic induction. Explain how this risk can be reduced. Write the names of any TWO applications of this phenomenon.

OR

b. Calculate the total resistance and the amount of current passing through the given circuit.



END OF PAPER

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